

Well Probe V2

Mechanical float switch for dry run protection of LORENTZ solar pumps

The well probe provides a reliable method of run dry protection for LORENTZ pumps. The well probe detects that water is present within a well, tank or other water source. The well probe is typically attached to the riser pipe above the pump and connected to the controller. When the well probe becomes dry (water level is below the probe) the pump switches off to avoid dry running.

Order Information

Item no.: 19-000005 **Product name:** Well probe sensor V2

Features

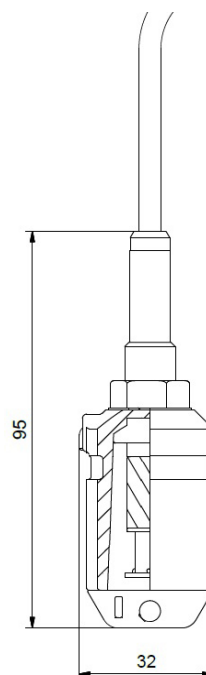
- Reliable dry run protection
- Simple to install using 3 cable ties
- Improved tolerance to dirt
- Splicing kit and cable ties for fixing are included

Technical Data

- Max. operating temperature 55°C
- Enclosure class: IP68
- Submersion depth: max 50 m (164 ft)
- Cable length: 1.5 m
- Wire size: 2 x 0.50 mm² or AWG 20, waterproofed
- Must be mounted in a vertical position
- Meets the requirements for CE

Dimensions / Weight

- Packaging dimensions: 255 x 170 x 40 mm
10.0 x 6.7 x 1.6 in
- Total weight: 0.1 kg / 0.2 lbs



Float Switch

Mechanically Activated Device for Water Level Detection in Applications with LORENTZ Solar Pump Systems

The switch can be used to detect the water level within a tank. When the water level in the tank reaches the maximum, the LORENTZ Controller will stop the pump and indicates Tank Full LED.

ORDER INFORMATION

- Item no.: 19-000030 product name: Float Switch

FEATURES

- **N.O.** (normally open) and **N.C.** (normally closed) function
- Reliable water level detection
- Simple to install
- Trouble free operation
- Not sensitive to rotation
- Corrosion-free

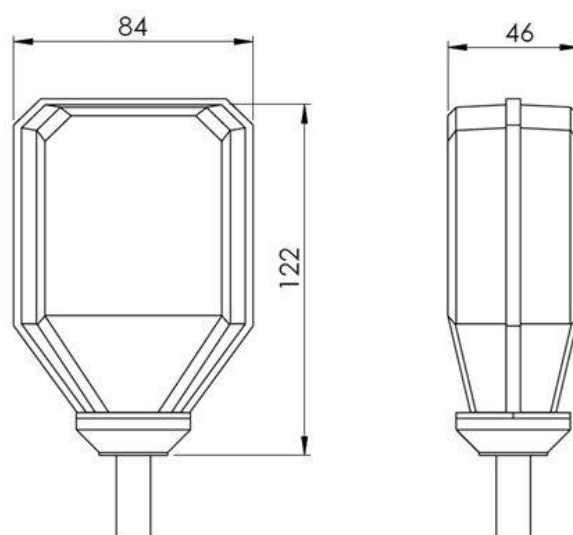


TECHNICAL DATA

- Operating temperature: -10°C to 55°C
- Storage temperature: : -10°C to 55°C
- Enclosure class: IP68
- Cable length: 3m, waterproof
- Wire size: 3x 1.0mm² or AWG 18
- Meets the requirements for CE

DIMENSION/WEIGHT

- Packaging dimensions: 230 x 160 x 55 mm
9.1 x 6.3 x 2.2 in
- Total weight: 0.8 kg / 1.8 lbs



Pressure Switch

Device for Water Pressure Detection in Applications with LORENTZ Solar Pump Systems

The switch can be used to detect the water level within a tank. When the water level in the tank reaches the maximum, the pressure will increase and the LORENTZ Controller will stop the pump and indicates Tank Full LED.

ORDER INFORMATION

- **Item no.:** 19-000310 **product name:** Pressure Switch 1-5bar

FEATURES

- Reliable water pressure detection
- Simple to install
- Trouble free operation
- Corrosion-free

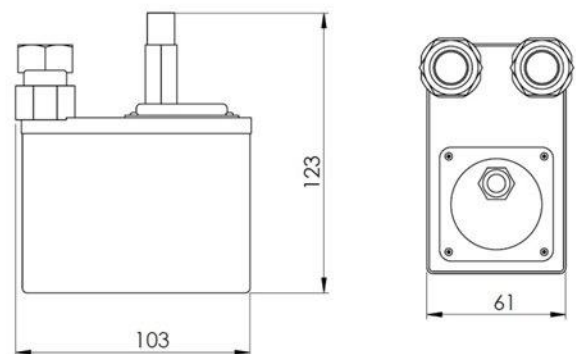
TECHNICAL DATA

- Pressure range between 1 and 5 bar
- 2x cable gland Ø 5-9mm
- G 1/4" pipe thread
- Meets the requirements for CE



DIMENSION/WEIGHT

- Packaging dimensions: 130 x 105 x 65 mm
5.1 x 4.2 x 2.6 in
- Total weight: 0.5 kg / 1 lbs



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Liquid Level Sensor

Sensor for measuring the level of liquid in a well or tank

USE / PURPOSE

The range of liquid level sensors use pressure to measure the liquid level in a well or tank. The sensors can be used for collection of long term liquid level monitoring and also for pump control in applications where a well probe cannot be used. The liquid level sensor must be used with a compatible LORENTZ Controller (see requirements).

FEATURES

- Gauge pressure sensor, pressure measurement relative to atmosphere
- For measuring levels of liquid in a well or tank
- Accurate, robust sensor
- For use with LORENTZ PumpScanner and pumpMANAGER

REQUIREMENTS

- LORENTZ PS2 controller, LORENTZ PSk2 controller or PS Controller equipped with a licensed PS DataModule
- The cable contains a capillary tube to balance the sensor against atmospheric changes. The sensor must be ordered with the required cable length for the application as it cannot be spliced. Further cable lengths are available on demand, please contact LORENTZ.

TECHNICAL DATA

- Sensor type: 2 wire gauge sensor
- Enclosure class: IP68
- Sensor housing: stainless steel
- Connects to LORENTZ PS DataModule
- Oil resistant cable with capillary tube
- Overpressure: 2x full scale
- Output signal :4-20mA output
- Voltage : 12-28VDC
- Application temp.: -10 to 80°C
14 to 176°F
- Accuracy class :0.5% FS
- Meets the requirements for CE

DIMENSION/WEIGHT

- Packing dimensions:
19-005040: 350 x 350 x 100 mm / 13.8 x 13.8 x 3.9 in
19-005050: 350 x 350 x 100 mm / 13.8 x 13.8 x 3.9 in
19-005080: 350 x 350 x 100 mm / 13.8 x 13.8 x 3.9 in
19-004395: 350 x 350 x 180 mm / 13.8 x 13.8 x 7.1 in
- Weight:
19-005040: 4 kg / 8.8 lbs
19-005050: 5 kg / 11.0 lbs
19-005080: 7 kg / 15.4 lbs
19-004395: 11 kg / 24.3 lbs



ORDER INFORMATION

Item #	Product	Pressure range	Cable length
19-005040	Liquid Level Sensor, 0-10m/33ft, 30m/100ft Cable	0-100 kPa 0 to 10 m / 0 to 33 ft 0 to 1 bar / 0 to 14.5 psi	30 m / 100 ft *
19-005050	Liquid Level Sensor, 0-20m/66ft, 40m/130ft Cable	0-200 kPa 0 to 20 m / 0 to 66 ft 0 to 2 bar / 0 to 29 psi	40 m / 130 ft *
19-005080	Liquid Level Sensor, 0-50m/164ft, 60m/200ft Cable	0-500 kPa 0 to 50 m / 0 to 164 ft 0 to 5 bar / 0 to 72.5 psi	60 m / 200 ft *
19-004395	Liquid Level Sensor, 0-100m/328ft, 140m/460ft Cable	0-1000 kPa 0 to 100 m / 0 to 328 ft 0 to 10 bar / 0 to 145 psi	140 m / 460 ft *

*Further cable length is available on demand. Please contact LORENTZ.

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Liquid Pressure Sensor

Sensor for measuring the pressure of liquid in a pipe or vessel

USE / PURPOSE

The sensors are commonly used to measure the pressure in the delivery pipeline or in a vessel. The pressure signal is used with a suitable LORENTZ pump controller to measure pressure and for pressure switching or constant pressure applications. The liquid pressure sensor must be used with a compatible LORENTZ Controller (see requirements).

FEATURES

- Gauge pressure sensor, pressure measurement relative to atmosphere
- For measuring pressure in a pipe or vessel
- Accurate, robust sensor
- For use with LORENTZ PumpScanner and pumpMANAGER

REQUIREMENTS

- LORENTZ PS2 controller, LORENTZ PSk2 controller or PS Controller equipped with a licensed PS DataModule
- Care must be taken to position the sensor without turbulent water to ensure accurate measurement
- G1/4" or G1/2" female threaded filling / air vent hole is required to mount the sensor



TECHNICAL DATA

- Sensor type: 2 wire gauge sensor
- Enclosure class: IP65
- Sensor housing : stainless steel
- Connects to LORENTZ PS DataModule
- 5m/10m (16ft/33ft) cable length
- Overpressure: 1.5x full scale
- Output signal : 4-20 mA
- Voltage : 11-28 VDC
- Application temp.: -30 to 80°C
-22 to 176°F
- Accuracy class : 0.5% full scale
- Thread type: G1/2" male (G1/4" with adapter)
- Meets the requirements for CS

ORDER INFORMATION

Item #	Product	Pressure range
19-004450	Liquid Pressure Sensor, LPS-500	0-500 kPa 0 to 5 bar / 0 to 72.5 psi
19-004460	Liquid Pressure Sensor, LPS-1000	0-1000 kPa 0 to 10 bar / 0 to 145 psi
19-002760	Liquid Pressure Sensor, LPS-2000	0-2000 kPa 0 to 20 bar / 0 to 290 psi

DIMENSION/WEIGHT

- Packing dimensions: 190 x 180 x 80 mm
7.5 x 7.1 x 3.2 in
- Weight: 0.6 kg / 1.3 lbs

Water Meters

Measuring instruments for water use

NWM Water Meter, WP Water Meter



Water meters should be used in cases where measuring the volume of water flow is relevant for an application. LORENTZ offers two types of water meters (NWM and WP) covering the pipe size range from DN25 to DN200.

NWM Water Meters are suitable for applications with a pipe size of DN25 and DN32, whereas WP Water Meters are suitable for a pipe size of DN50 up to DN200.

Each water meter comes with a pre-mounted reed switch output. This output can be connected to PS2 and PSk controllers. This connection allows for monitoring of the flow rate and the amount of water pumped using the LORENTZ CONNECTED apps.

This datasheet includes product information for both the NWM Water Meter and the WP Water Meter.



NWM Water Meter



WP Water Meter

NWM Water Meter



Features

The NWM (Ningbo Water Meter) Water Meter is suitable for applications with a pipe size from DN25 to DN32.

- Hermetically liquid sealed rollers which ensures clear reading in any operating conditions
- Low pressure loss, long working life
- Easy to install
- Reed switch output for easy water flow control and monitoring

Technical Data

- Water temperature 50°C
- Water pressure: max. 16 bar
- IP64
- CE Conformity



Order information

item number	description
19-002220	water meter, NWM-DN25, 0.1 cbm/p
19-002225	water meter, NWM-DN32, 0.1 cbm/p

Accuracy Curve

The Accuracy curve shows the deviation in percent for different flow rates. In regular operation the deviation is between -2% and +2%.

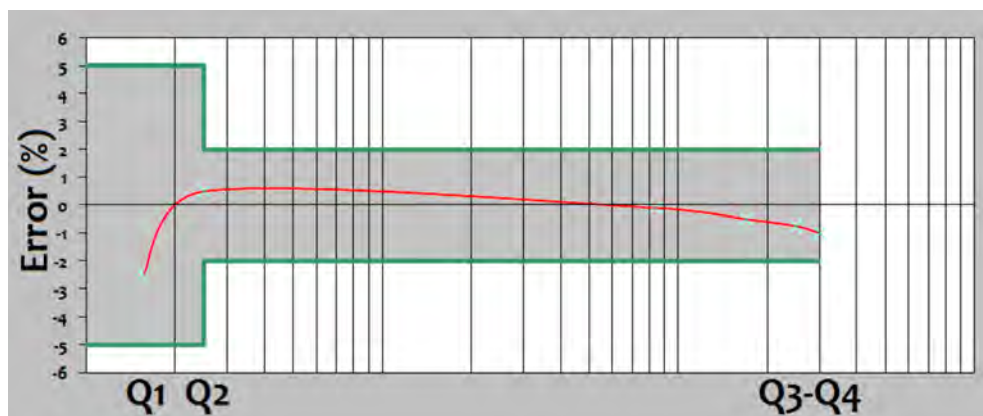
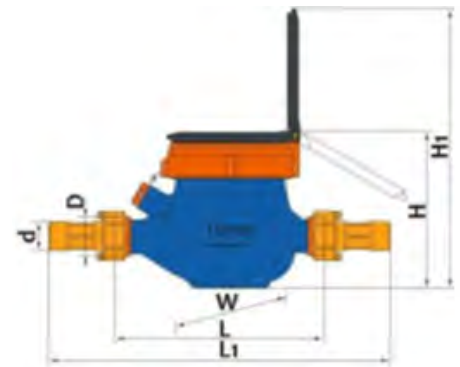


Table 1: NWM - Flow rate characteristics

	DN25	DN32
max. flow rate: $Q_{\max}$ [m³/h]	7.875	12.5
nominal flow rate: Q_n [m³/h]	0.063	0.1
transition flow rate: Q_t [m³/h]	3.0	5.0
minimum flow rate: $Q_{\min}$ [m³/h]	0.7	0.75

Table 2: NWM - Dimensions, weight specifications

	DN25	DN32
L	260	260
L1	380	384
D	G1-1/4B	G1-1/2B
d	R1	R1-1/4
H	117.5	117.5
H1	206.5	206.5
W	98	98



WP Water Meter



The WP (Woltman) Water Meter is suitable for applications with a pipe size from DN50 to DN200.

Features

- Dry dial register ensures clear reading
- Low pressure loss, long working life
- Easy to install
- Reed switch output for easy water flow control and monitoring

Technical Data

- Water temperature 40°C
- Water pressure: max. 16 bar
- IP64
- CE Conformity



Order information

item number	description
19-002165	water meter, WP-DN50, 0.1 cbm/p
19-002170	water meter, WP-DN65, 0.1 cbm/p
19-002180	water meter, WP-DN80, 0.1 cbm/p
19-002190	water meter, WP-DN100, 0.1 cbm/p
19-002200	water meter, WP-DN125, 0.1 cbm/p
19-002210	water meter, WP-DN150, 0.1 cbm/p
19-002202	water meter, WP-DN200, 0.1 cbm/p

Accuracy Curve

The Accuracy curve shows the deviation in percent for different flow rates. In regular operation the deviation is between -2% and +2%.

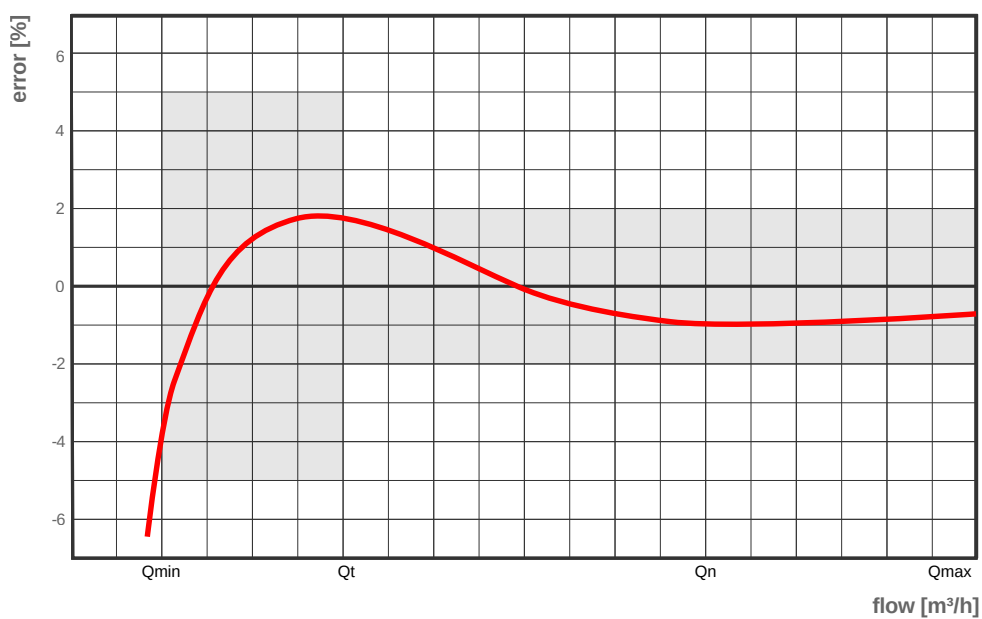
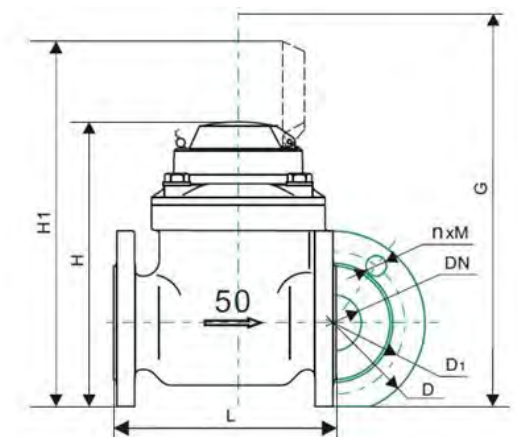


Table 1: WP - Flow rate characteristics

	DN50	DN65	DN80	DN100	DN125	DN150	DN200
max. flow rate: $Q_{\max}$ [m³/h]	30	50	80	120	200	300	500
nominal flow rate: Q_n [m³/h]	15	25	40	60	100	150	250
transition flow rate: Q_t [m³/h]	3.0	5.0	8.0	12	20	30	50
minimum flow rate: $Q_{\min}$ [m³/h]	0.7	0.75	1.2	1.8	3.0	4.5	7.5

Table 2: WP - Dimensions, weight specifications

	DN50	DN65	DN80	DN100	DN125	DN150	DN200
L [mm]	200	200	225	250	250	300	350
H [mm]	232	242	252	262	275	325	355
H1 [mm]	303	313	323	333	346	396	426
G [mm]	360	360	360	360	360	420	420
D [mm]	165	185	200	220	250	285	340
D [mm]	125	145	160	180	210	240	295
Connecting bolt quantity	4xM16	4xM16	8xM16	8xM16	8xM16	8xM20	12xM20
Weight [kg]	12	13	16	18	20	42	74



About LORENTZ

LORENTZ is the global market leader in solar powered water pumping solutions. Founded in Germany during 1993 LORENTZ has pioneered, innovated and excelled in the engineering and manufacturing of solar powered water pumping. Today LORENTZ is active in over 130 countries through a dedicated network of professional partners. LORENTZ technology uses the power of the sun to pump water, sustaining and enhancing the life of millions of people, their livestock and crops.

Simply – **Sun. Water. Life.**



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SunSwitch

Automatic Solar Level Switching Device

The SunSwitch can be used for any application where you need to switch depending on the solar intensity. It is an autonomous working unit that supplies itself with power from the PV-module on its rear side.

ORDER INFORMATION

- Item no.: 19-000050 **product name:** SUN SWITCH Light-Sensor

FEATURES

- Trouble free operation
- Pump will only run when minimum flow rates or pressure levels can be achieved which are required by certain applications
- Reduced wear on the pump and motor since the pump only starts when it will continue to run

TECHNICAL DATA

- Contact rating:
 - Max. voltage 250 VAC / 220 VDC
 - Max. current 2A
 - Max. power 60 W / 62.5 VA
- Enclosure class: IP54
- Cable length: 4 m
- Wire size: 3x 1.0mm² or AWG 18, waterproofed
- Mounted close to the solar array
- Manual with further information is available in PartnerNet
- Meets the requirements for CE

DIMENSION/WEIGHT

- Packaging dimensions: 230 x 115 x 90 mm
9.1 x 4.5 x 3.5 in
- Total weight: 1.2 kg / 0.6 lbs



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PV Disconnect 440-40-1

Connection box with DC disconnect

Description

An outdoor rated, combining connection box with DC disconnect switch that allows 1 strings of PV modules to be connected safely to a solar pump system.

The PV disconnect is also designed to accept an optional lightning protection device.

Features

- DC rated disconnect to provide safe isolation of the system
- Robust weather proof housing designed to make installation simple
- Lockable to secure the system during maintenance (power locked off)
- For professional installation of pumping systems
- Internal touch protection with screws
- Designed to be used with LORENTZ PS2-150 to PS2 4000 systems



photo may differ from actual product

Ordering and shipping information

- Item no: 19-000125
- Product name: PV Disconnect 440-40-1
- Packed volume 0.01 m³ (0.35 ft³)
- Packed weight 1.9 kg (4.2 lbs)

Approvals and standards

- Switch IEC 60947-3



Technical data / Specifications

Maximum voltage	440 V DC	
Maximum current per string	40 A	
Maximum total current	40 A	
Number of strings.	1	
Input cables	4 - 10 mm ²	AWG 12 - 8
Output cables	4 - 10 mm ²	AWG 12 - 8
PG glands (input)	2 x M16	
PG glands (output)	2 x M16	
Lightening protection mounting hole	PG16 cap	
Environmental protection	IP68	NEMA6
Housing material	Polycarbonate	

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Optional lightning surge protector

Provides protection for the pump controller from incoming high voltages on the PV side. The surge protector connects through a pre-drilled and blanked mounting hole in the PV connect housing

- Proper grounding of the device is required to achieve protection
- Item no.: 19-002120 MNSPD-115 PS2-150 to PS2-200
- Item no.: 19-002130 MNSPD-300 PS2-600 to PS2-1800
- Item no.: 19-002140 MNSPD-600 PS2-4000



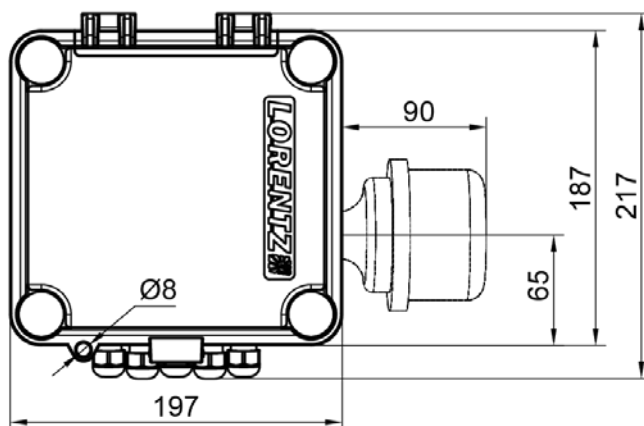
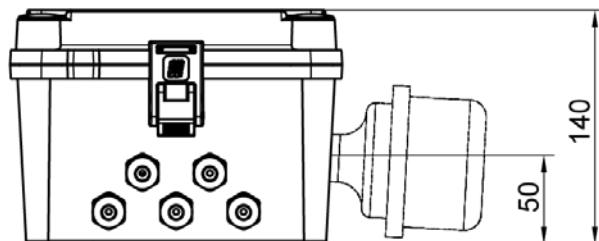
Mounting options

- Wall mount using 4 holes with weather protection
- Designed for optional pole mounting. Mounting points are pre-marked inside the housing.



Dimensions and weight

- See diagram for mm sizes
- Max height 220 mm (8.66")
- Max width (no surge protector) 197 mm (7.75")
- Max width (surge protector) 297 mm (Max 11.7")
- depth 140 mm (5.5")
- Weight 1403 g (3.1 lbs)



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PV Disconnect 440-40-3

Connection box with DC disconnect

Description

An outdoor rated, combining connection box with DC disconnect switch that allows up to 3 strings of PV modules to be connected safely to a solar pump system.

The PV disconnect is also designed to accept an optional lightning protection device.

Features

- DC rated disconnect to provide safe isolation of the system
- Combining function, allows 3 strings to be connected to the solar pump system
- Robust weather proof housing designed to make installation simple
- Lockable to secure the system during maintenance (power locked off)
- For professional installation of pumping systems
- Internal touch protection with screws
- Designed to be used with LORENTZ PS2-150 to PS2 4000 systems

Ordering and shipping information

- Item no: 19-000126
- Product name: PV Disconnect 440-40-3
- Packed volume: 0.01 m³ (0.35 ft³)
- Packed weight: 2.1 kg (4.6 lbs)

Approvals and standards

- Switch IEC 60947-3



photo may differ from actual product

Technical data / Specifications

Maximum voltage	440 V DC	
Maximum current per string	15 A	
Maximum total current	40 A	
Number of strings.	3	
Input cables	2.5 - 4 mm ²	AWG 14 - 12
Output cables	4 - 10 mm ²	AWG 12 - 8
PG glands (input)	6 x M16	
PG glands (output)	2 x PG11	
Lightening protection mounting hole	PG16 cap	
Environmental protection	IP68	NEMA6
Housing material	Polycarbonate	

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Optional lightning surge protector

Provides protection for the pump controller from incoming high voltages on the PV side. The surge protector connects through a pre-drilled and blanked mounting hole in the PV connect housing

- Proper grounding of the device is required to achieve protection
- Item no.: 19-002120 MNSPD-115 PS2-150 to PS2-200
- Item no.: 19-002130 MNSPD-300 PS2-600 to PS2-1800
- Item no.: 19-002140 MNSPD-600 PS2-4000



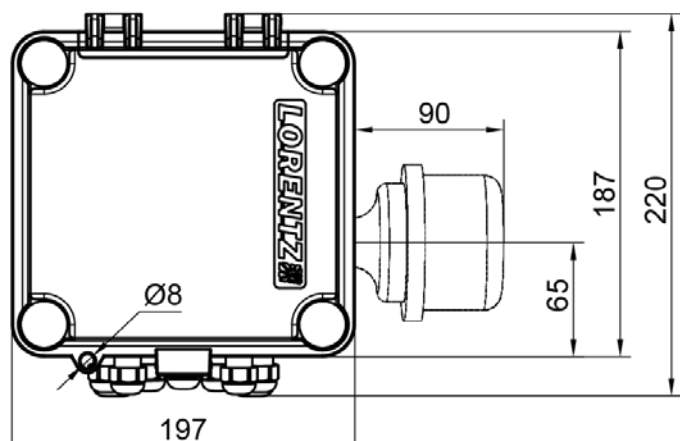
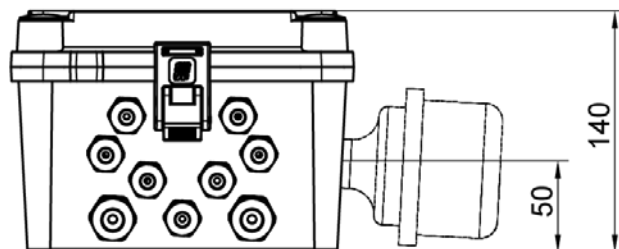
Mounting options

- Wall mount using 4 holes with weather protection
- Designed for optional pole mounting. Mounting points are pre-marked inside the housing.



Dimensions and weight

- See diagram for mm sizes
- Max height 220 mm (8.66")
- Max width (no surge protector) 197 mm (7.75")
- Max width (surge protector) 297 mm (Max 11.7")
- depth 140 mm (5.5")
- Weight 1.62 kg (3.56 lbs)



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PV Disconnect 440-40-6

Connection box with DC disconnect

Description

An outdoor rated, combining connection box with DC disconnect switch that allows up to 6 strings of PV modules to be connected safely to a solar pump system.

The PV disconnect is also designed to accept an optional lightning protection device.

Features

- DC rated disconnect to provide safe isolation of the system
- Combining function, allows 6 strings to be connected to the solar pump system
- Robust weather proof housing designed to make installation simple
- Lockable to secure the system during maintenance (power locked off)
- For professional installation of pumping systems
- Internal touch protection with screws
- Designed to be used with LORENTZ PS2-150 to PS2 4000 systems

Ordering and shipping information

- Item no: 19-000124
- Product name: PV Disconnect 440-40-6
- Packed volume: 0.015 m³ (0.54 ft³)
- Packed weight: 2.7 kg (6 lbs)

Approvals and standards

- Switch IEC 60947-3



photo may differ from actual product

Technical data / Specifications

Maximum voltage	440 V DC	
Maximum current per string	10 A	
Maximum total current	40 A	
Number of strings.	6	
Input cables	2.5 - 4 mm ²	AWG 14 - 12
Output cables	4 - 10 mm ²	AWG 12 - 8
PG glands (input)	12 x M16	
PG glands (output)	2 x PG11	
Lightening protection mounting hole	PG16 cap	
Environmental protection	IP68	NEMA6
Housing material	Polycarbonate	

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Optional lightning surge protector

Provides protection for the pump controller from incoming high voltages on the PV side. The surge protector connects through a pre-drilled and blanked mounting hole in the PV connect housing

- Proper grounding of the device is required to achieve protection
- Item no.: 19-002120 MNSPD-115 PS2-150 to PS2-200
- Item no.: 19-002130 MNSPD-300 PS2-600 to PS2-1800
- Item no.: 19-002140 MNSPD-600 PS2-4000



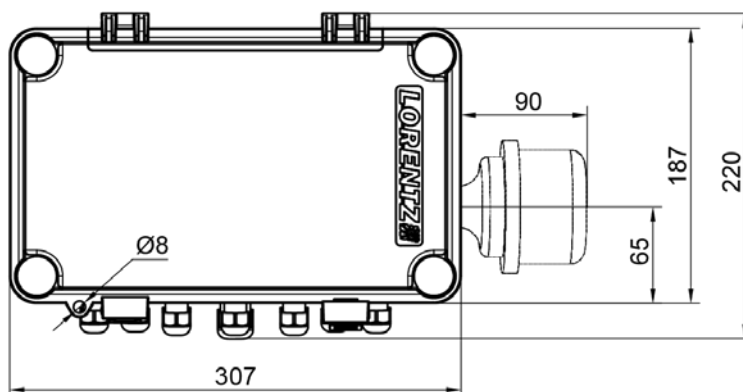
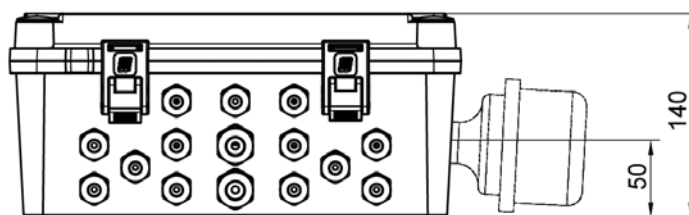
Mounting options

- Wall mount using 4 holes with weather protection
- Designed for optional pole mounting. Mounting points are pre-marked inside the housing.



Dimensions and weight

- See diagram for mm sizes
- Max height 220 mm (8.66")
- Max width (no surge protector) 307 mm (12.1")
- Max width (surge protector) 397 mm (Max 15.6")
- depth 140 mm (5.5")
- Weight 2.13 kg (4.7 lbs)



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Sacrificial Anode for HR Pump

Zinc sacrificial anode for corrosion protection of HR pump ends

USE / PURPOSE

The LORENTZ Sacrificial Anode is designed for use with LORENTZ pump ends. Where solar pump ends operate in aggressive water conditions acidic water can attacks the passive layer of the high quality-steel pump end. This corrosion reduces the life time of the product. The LORENTZ Sacrificial Anode protects the pump system from this type of corrosion and extends the pump life

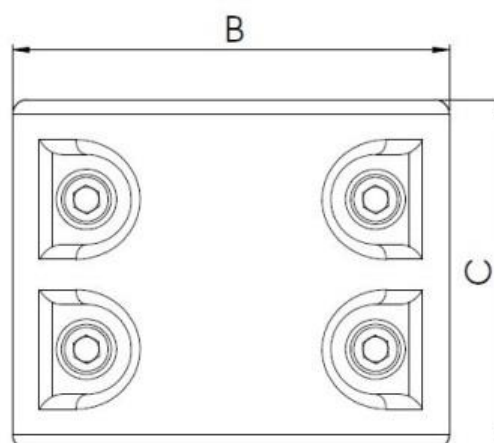
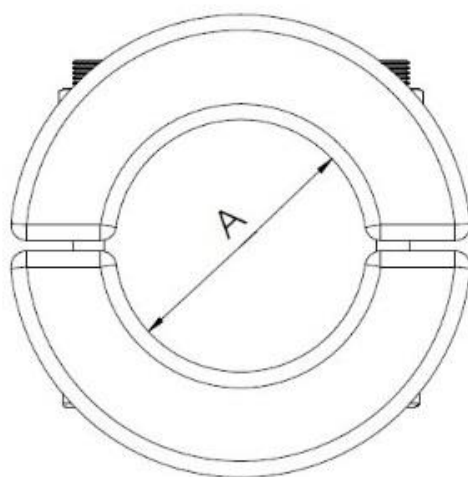
ORDER INFORMATION

- Item no.: 19-002621 Product name: Anode, Sacrificial Zink Anode, Di42mm
- Item no.: 19-002622 Product name: Anode, Sacrificial Zink Anode, Di48mm

FEATURES

- Simple and cost effective part to protect the pump end from corrosion
- Designed for HR pump ends

TECHNICAL DATA



	A	B	C	Weight	Surface	Item no.
HR03/04/05	48 mm 1,89 in	88 mm 3,46 in	70 mm 2,76 in	1,76 kg 3,88 lbs	59100 mm ² 91,61 in ²	19-002622
HR07/10/14/23/32	42 mm 1,65 in	82 mm 3,23 in		1,68 kg 3,70 lbs	55500 mm ² 86,01 in ²	19-002621

PP600H AC PowerPack

AC/DC Converter to Supply PS2-600 and specific PS2-1800HR Pump Systems with Backup Power from a Generator or Mains Supply

ORDER INFORMATION

- Item no.: 19-001040 product name: PowerPack 600H, UL

TECHNICAL DATA

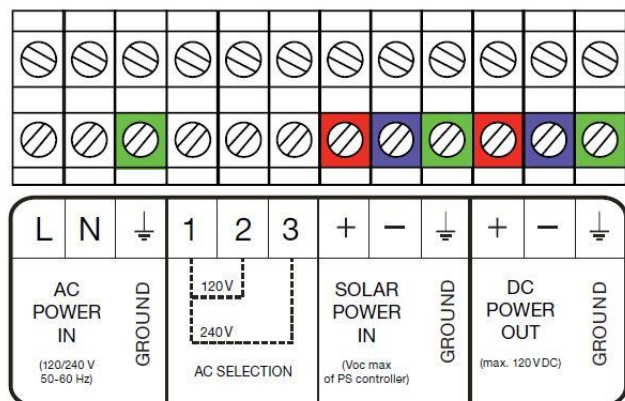
- AC input: 120V or 240V ($\pm 15\%$), selectable, 47 to 63Hz
- AC input current: 6.2A (120V) or 3.1A (240V)
- PV input voltage: V_{mp} : 130V DC, depending on pump controller max. voltages
- PV max open circuit voltage: 200V DC depending on pump controller max. voltages
- DC output: 120V DC (no load), 700W
- Overload protection: internal fuse, 15A
- Enclosure: steel, gasket-sealed, indoor-use, hinged front cover with key-lock
- Must be protected from direct mid-day sun
- Enclosure class: IP22
- Ambient temperature: max. 45°C
- All wired must be #14 AWG (2.5 mm²) or larger
- Manual with further information is available in partnerNET
- Meets requirements for CE

Electrical Safety



DIMENSION/WEIGHT

- 450 x 300 x 140 mm
17.7 x 11.8 x 5.5 in
- Total weight: 16 kg / 35 lbs



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PM Accessory bracket

Kit for mounting LORENTZ PS Controller / PV-Disconnect to the Pole Mount PM-1-X

USE / PURPOSE

The PM Accessory bracket is designed for use with the LORENTZ POLE MOUNT-1-X.

The Set allows an easy and flexible mounting of LORENTZ PS Controllers and PV-Disconnect on the Pole of the LORENTZ PM-1-X.



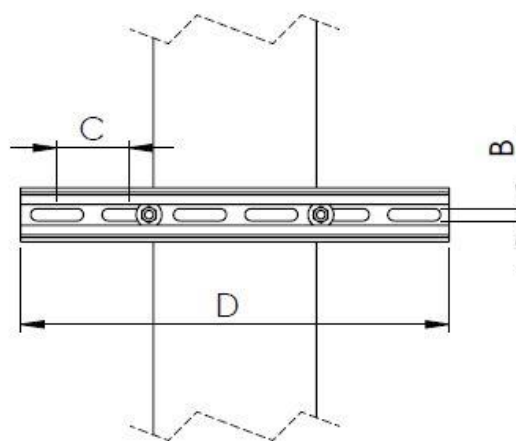
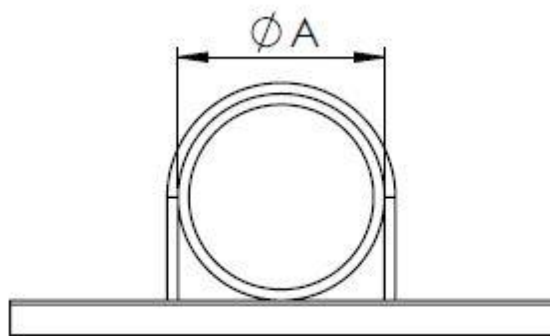
ORDER INFORMATION

Part	Item No.	PM-1-1	PM-1-2	PM-1-3	PM-1-4	PM-1-5	PM-1-6
PM Accessory bracket 2.5-1	19-004220	x	x				
PM Accessory bracket 4-1	19-004210			x	x	x	x

FEATURES

- Simple and cost effective mounting possibility for LORENTZ Controller and PV-Disconnect
- Designed for LORENTZ Pole Mount 1-X

TECHNICAL DATA



	A	B	C	D	Weight	Item no.
PM-1- 1/2	70 -73 mm 2.5 " SCH40	8.5 x 37 mm 0.33 x 1.4 "	50 mm 2 "	300 mm 11.8 "	0.35 kg 0.77 lbs.	19-004220
PM-1- 3/4/5/6	114.3 mm 4 " SCH40				0.4 kg 0.88 lbs.	19-004210

Packing dimensions

305x170x25 mm
12x7x1 inch

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Sun Sensor

Irradiation measurement module and software

The LORENTZ Sun Sensor module and software allow the PS2 and PSk2 controller to accurately estimate irradiation levels and control the pump based on this.

The Sun Sensor function can be configured to offer the pump mechanical protection from repeated starts and stops due to low irradiation. The Sun Sensor function can also be configured to switch external devices via the output terminal.

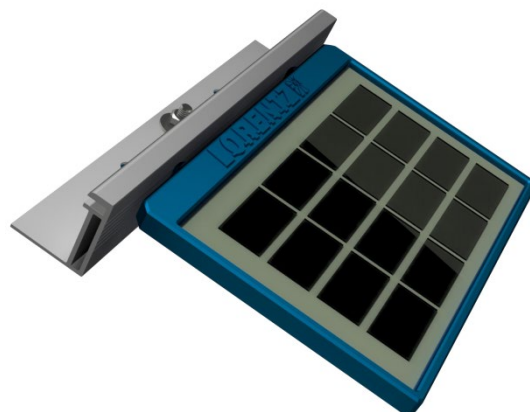
Each module is tested and specially calibrated for use with the PS2 and PSk2 system. The module also includes a software license to use the features which is accessed via the item serial number.

This Sun Sensor module is included with any PS2-1800 or PS2-4000 controller and any PSk2 controller.

For PS controllers (non PS2) refer to the SunSwitch Datasheet.

ORDER INFORMATION

- **Item no.:** 19-005030
- **Product name:** Sun Sensor module



TECHNICAL DATA

- Open circuit voltage $V_{oc} = 10.2 \text{ V}$
- Short circuit current $I_{sc} = 0.19 \text{ A}$
- Wire size: $2 \times 1.0 \text{ mm}^2$ / AWG 18, waterproofed
- Mount in parallel to the solar array
- PS2/PSk2 manual with further information is available in partnerNet
- Meets the requirements for CE

DIMENSION/WEIGHT

- Packaging dimensions: 180 x 205 x 60 mm
7.08 x 8.07 x 2.36 in
- Total weight: 0.6 kg / 1.33 lbs

Flow Sleeve for HR(E) Pumps

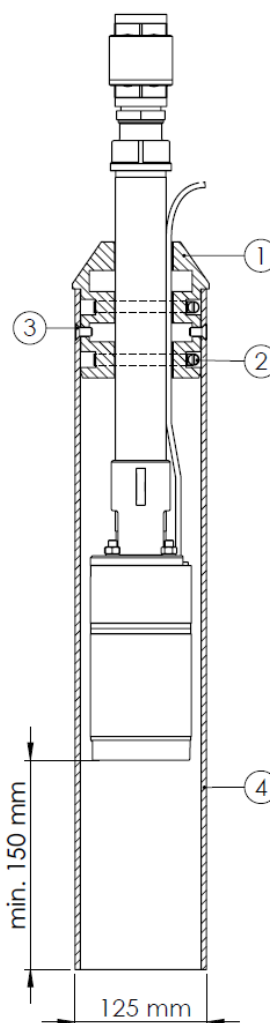
Device to protect the pump from dirt and for motor cooling

A flow sleeve is a sleeve that goes around the pump body and motor. It has two functions:

1. **A flow sleeve reduces the amount of dirt in the pump.** Where water sources contain sand or have the potential for dirt falling into them (uncapped wells) a flow sleeve is recommended to protect the pump from dirt. Due to the large inner diameter of the flow sleeve the water rises very slowly and dirt deposits at the bottom of the well. Subsequently, water with less sediments enters the pump which increases its lifetime.
2. **A flow sleeve cools the motor.** In open water, tanks or pools there is no natural flow of water across the motor which can lead to heat build-up. The tube enforces water flow across the motor which has a cooling effect.

ORDER INFORMATION (please compare (1), (2) etc. with the sample flow sleeve)

Item no.	Product name	Scope of delivery	Suitable for
19-000220	Flow Sleeve Kit for 4" HR Pumps	(1) 1 x flow sleeve adapter (2) 2 x hose clamp (3) 4 x countersunk bolt M6x20 (4) 1 x PVC tube: $\varnothing 125 \times 5$, 800 mm length	All HR pumps with 4" motor
19-000225	Flow Sleeve Kit for 4" HRE Pumps		All HRE pumps with 4" motor
19-000240	Flow Sleeve Adaptor for 4" HR Pumps	(1) 1 x flow sleeve adaptor (2) 2 x hose clamp (3) 4 x countersunk bolt M6x20	All HR pumps with 4" motor
19-000245	Flow Sleeve Adaptor for 4" HRE Pumps	A tube (4) is not part of the scope of delivery.	All HRE pumps with 4" motor



sample flow sleeve

FEATURES

- Due to the large inner diameter of the flow sleeve the water rises very slowly allowing dirt to settle at the bottom of the well
- The life expectancy of the pump is increased due to less sand entering the pump
- Simple installation
- Can be used with PS2-200 – PS2-4000 HR(E) Pumps
- Min. well diameter of 6" (150 mm) required

DIMENSIONS/WEIGHT

Flow Sleeve Kit for 4" HR(E) Pumps (19-000220 or 19-000225):

- Packaging dimensions: 850 x 150 x 130 mm
33.5 x 6.0 x 5.1 in
- Total weight: 3.7 kg / 8.2 lbs

Flow Sleeve Adaptor for 4" HR(E) Pumps (19-000240 or 19-000245, without PVC tube):

- Packaging dimensions: 135 x 135 x 135 mm
5.3 x 5.3 x 5.3 in
- Total weight: 1.2 kg / 2.6 lbs

PS2 Manual Speed Controller

Device to provide manual motor speed control of PS2 systems

The LORENTZ PS2 Manual Speed Controller allows adjustment of the maximum motor speed without using the PumpScanner App. To use the Manual Speed Controller, it is required to activate this function in the settings of PumpScanner during or before installation.



ORDER INFORMATION

- Item no.: 19-000035 Product name: PS2 Manual Speed Controller

FEATURES

- Allows manual control of PS2 motor speed
- Outdoor rated, installed in the housing of the controller

TECHNICAL DATA

- Voltage: 15-24 V DC
- Enclosure class: IP65
- Ambient temperature: -38...50 °C (-36... 122 F)
- Wire size: 2 x 0,75 mm²/18 AWG
- Replaces Ø 20mm cable gland
- Meets the requirements for CE
- Please note that if "Manual Speed Controller" is configured then "Set speed limitation" function is not available in PumpScanner.

DIMENSION/WEIGHT

- Packing dimension : 100 x 70 x 35 mm; 3.9 x 2.7 x 1.3 in
- Total weight: 0.2 kg / 0.4 lbs



PS 2 Controller Plug Kit

Kit for an easy and electrical safe installation of PS2 Controller

The LORENTZ PS2 Controller Plug Kit can be installed on any PS2 system. The kit extends the internal wiring connections to plugs allowing systems to be pre-wired and delivered to site. Possible uses for the plug kit are where time on site needs to be minimized, where systems are often moved or where the skills available onsite do not allow for a standard installation.

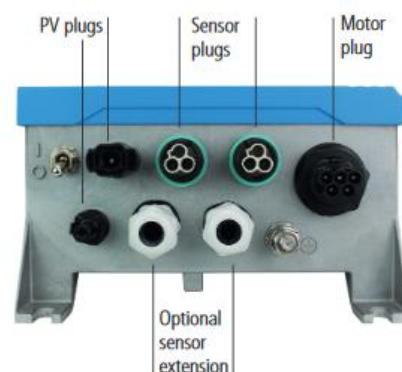
The sensor extension kit allows the installation of two additional sensors, the standard set contains only two plug sets for sensor connections.

Order Information

- 19-005001 Plug- Kit PS2- Controller
- 19-005011 Sensor Plug Extension Kit

Features

- Allows fast, easy and electrical safe installation of PS2 Controller
- Customer must not open the controller for installation
- Outdoor rated, all parts are designed for outdoor use



Technical Data

PLUG	Wire size	max. current	max. voltage	Ambient temperature
Motor	max, 6mm ² (10 AWG)	32 A	600V DC	-40°C ... +90°C
PV	max 8mm ² (8 AWG)	40 A (27A at 2.5mm ² /14AWG)	1500V DC	-40°C ...+90°C
Sensor	max. 1.5mm ² (16 AWG)	3 A	50V DC	-40°C ...+90°C

Packing Dimension/Weight

19-005001 Plug- Kit PS2- Controller

- Packing dimension: 16 x 300 x 4cm (6.3x12x1.6inch)
- Total weight: 0.33kg (0,73lb)

19-005011 Sensor Plug Extension Kit

- Packing dimension: 16 x 24 x 2cm (6.3x10x0.8inch)
- Total weight: 0.1kg (0,22lb)



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Surge Protector2

Device to Protect LORENTZ Pump Accessories from Voltage Spikes

ORDER INFORMATION

- Item no.: 19-005210 product name: Surge Protector2

FEATURES/COMPATIBILITY

- Reliable surge protection device for any switched, pulse or analogue (4-20 mA) inputs sensors including:
 - Well Probe Sensor 19-000000
 - Water Sensor 19-000001
 - Float Switch 19-000030
 - Pressure Switch 19-000310
 - Liquid Level (all types, e.g. 19-005040)
 - Liquid Pressure Sensor (all types, e.g. 19-004460)
 - Water Meter (all types, e.g. 19-002160)
 - Sun Switch (19-000050)
- The device must be installed inside the PS2 or PSk2 controller.



TECHNICAL DATA

- Max. voltage: 30 V DC
- Max current 8/20μs: 500 A
- Enclosure class: IP20
- Ambient temperature: max. 80°C (176°F)
- Wire size: 2 x 1.5mm² (AWG 16)
- Meets the requirements for CE

DIMENSION/WEIGHT

- Packing dimensions: 56 x 26 x 120 mm
 2.2 x 1.02 x 0.47 in
- Total weight 0.1 kg / 0.2 lbs

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PowerPack 600S

DC power supply for PS2 pump systems

Description

An outdoor rated power supply unit to allow LORENTZ PS2-200 to PS2-600 systems to be powered from an AC (mains voltage) power source. The PowerPack can be used as a continuous supply, in conjunction with PV modules or as a temporary or seasonal supply.

Features

- DC power supply to power LORENTZ solar pump systems from the grid or a generator
- Robust weatherproof housing
- External on / off switch
- Use of MC4 connections to allow quick change over from PV modules to PowerPack
- DC inputs and outputs allow for permanent connection of a solar array
- 1m AC connection cable - NEMA 5-15P (US Plug) or CEE 7/7 (EU Plug)
- Requires a tool to open the housing
- Designed to be used with the LORENTZ PS2-100 to PS2-600 systems

Ordering and shipping information

- Packed volume: 0.03 m³ (1.03 ft³)
- Packed weight: 3.6 kg (7.1 lbs)
- See next page for item numbers and variants

Approvals and standards

The switch mode power supply has the following approvals

- EN62368-1
- IEC 62103
- IEC 61000-6-2
- IEC 61000-6-4
- AS/NZS 60950-1
- UL62368-1



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Technical data / Specifications

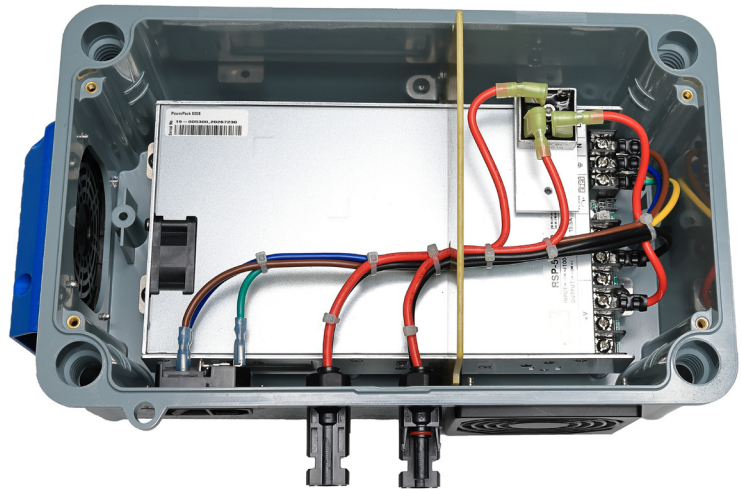
PowerPack 600S	
For use with these Pump systems	PS2-200 PS2-600
AC input	100-260 V AC / 50 - 60 Hz
DC output	56 V - 10.5 A
Maximum PV input	150 V DC
PV in connections	2 x MC4
DC out connections	2 x MC4
AC input	IEC C14 female (US or EU cable supplied)
Environmental protection	IP23
Housing material	Polycarbonate

Order information

Item number	Description
19-005300	PowerPack 600S, EU
19-005305	PowerPack 600S, US

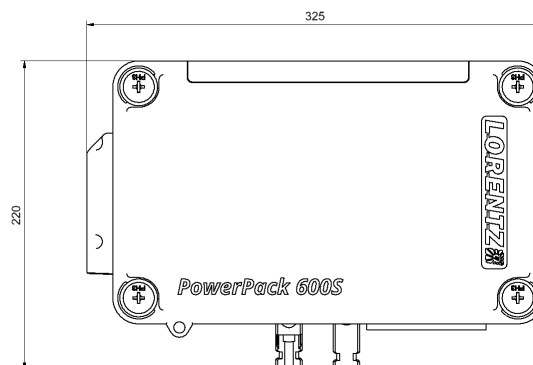
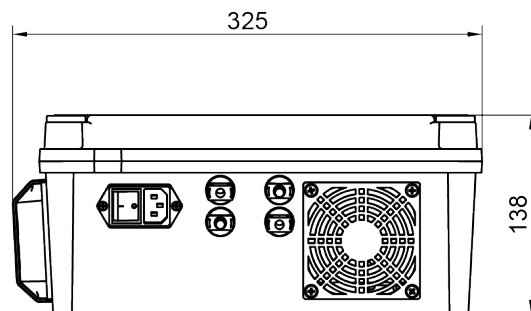
Mounting options

- Wall or panel mount using 4 holes with weather protection



Dimensions and weight

- See drawing for mm measurements
- Max height 220 mm (8.7")
- Max width 325 mm (12.8")
- depth 138 mm (5.4")
- Weight 2.9 kg (6.4 lbs)



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